



SWIFT-KD: Sliding Window Intelligent Federated Transformer Learning with Knowledge Distillation for Building Energy Prediction

Jessica Al Achy¹ · Hassan Harb² · Abdallah Makhoul¹

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Abstract

Building energy forecasting plays a crucial role in improving energy efficiency and sustainability, yet large-scale deployment faces critical challenges from privacy regulations, data heterogeneity, and communication constraints. Traditional Federated Learning (FL) approaches struggle with two fundamental limitations: inability to efficiently process extended temporal sequences and prohibitive communication overhead on resource-constrained edge devices. This paper introduces SWIFT-KD, a federated learning framework that addresses these challenges through hierarchical sliding window transformers and federated knowledge distillation. The hierarchical architecture decomposes long energy sequences into overlapping segments, capturing both fine-grained local patterns and long-term dependencies while maintaining computational efficiency for edge deployment. Knowledge distillation (KD) compresses model updates by transmitting soft predictions instead of full weights, achieving a 300-fold communication reduction. Evaluation on the ASHRAE dataset with 100 heterogeneous buildings demonstrates superior performance (Coefficient of Determination (R^2) = 0.9708, Root Mean Squared Error (RMSE) = 92.24 kWh, Mean Absolute Error (MAE) = 44.58 kWh), outperforming the standard federated averaging by 21% and exceeding the centralized training by 13.3%. The framework reaches peak performance within two communication rounds and maintains stable accuracy throughout the remaining rounds of training. These results establish that privacy-preserving distributed learning can achieve the accuracy requirements of practical building management without sacrificing efficiency.

Keywords Federated learning · Knowledge distillation · Transformer models · Energy forecasting · Smart buildings · Deep learning

Hassan Harb and Abdallah Makhoul contributed equally to this work.

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Extended author information available on the last page of the article

1 Introduction

Accurate energy forecasting plays a key role in improving the efficiency and sustainability of smart buildings (Harb and Makhoul 2019). Although data-driven models have shown strong potential in capturing complex consumption patterns, their large-scale deployment is often limited by data privacy concerns, system heterogeneity, and resource constraints at the edge (Harb et al. 2019; Makhoul et al. 2015). Federated learning offers a promising alternative by enabling collaborative model training without sharing raw data, yet existing solutions remain inefficient in processing long temporal sequences and minimizing communication overhead. This paper addresses these limitations by proposing an efficient federated framework for distributed building energy forecasting.

1.1 Context and Background

Buildings account for approximately 40% of global energy consumption, with demand growing by 3–4% annually (IEA: Buildings 2022). Recent advances in deep learning have revolutionized energy forecasting, yet large-scale deployment faces critical challenges: privacy regulations prevent centralized data aggregation across facilities (GDPR 2022), and heterogeneity in building types results in diverse data distributions that complicate model generalization (Xiao et al. 2023). Federated learning addresses these challenges by enabling collaborative model training without direct data sharing (McMahan et al. 2017). However, existing implementations struggle with degraded performance under non-IID data distributions, insufficient temporal modeling capabilities, and substantial communication overhead on resource-constrained edge devices (Liu et al. 2020). Although recent transformer architectures show promise in capturing temporal dependencies (Zhou et al. 2021), they lack mechanisms to handle very long sequences efficiently and generate prohibitively large model updates for federated deployment.

1.2 Motivation and Challenges

Current federated learning approaches for building energy forecasting face interconnected limitations. Heterogeneous data distributions across buildings can degrade prediction accuracy by up to 55% with conventional Federated Averaging (FedAvg) aggregation (Zhao et al. 2018). Traditional architectures such as Long Short-Term Memory (LSTM) and Convolutional Neural Network (CNN) (Rafi et al. 2021; Kim et al. 2025) fail to adequately capture long-range temporal dependencies, while standard transformers struggle to process extended energy consumption sequences due to quadratic computational complexity (Zhou et al. 2021). Processing very long historical data with attention mechanisms becomes prohibitively expensive and suffers from information dilution, where relevant patterns become obscured over extended time periods (Li et al. 2022).

Communication overhead remains a significant challenge because transmitting full model weights during each federated round requires substantial bandwidth and energy. Furthermore, conventional aggregation methods assign equal weights to all client updates regardless of their quality (Jithish et al. 2023), thereby potentially reducing the accuracy and robustness of the global model. Two key opportunities remain unexplored: integrating hierarchical sliding-window mechanisms to efficiently process long sequences by decom-

posing them into manageable overlapping segments, and applying knowledge distillation to compress model updates and reduce communication payloads without sacrificing performance. SWIFT-KD addresses these gaps by extending federated learning architectures with efficient temporal modeling and communication protocols specifically designed for distributed building energy forecasting.

1.3 Main Contributions

This paper proposes SWIFT-KD, a federated learning framework for building energy forecasting in an edge–fog–cloud architecture. SWIFT-KD incorporates two key innovations. First, it introduces a hierarchical sliding window transformer that partitions long energy consumption sequences into overlapping windows. Local transformers learn fine-grained temporal patterns from each window, while a global transformer captures long-term dependencies by integrating local representations. This hierarchical design preserves both local and global temporal information with moderate computational complexity. Second, SWIFT-KD employs a federated knowledge distillation protocol to reduce communication overhead. Instead of transmitting full model parameters, edge devices exchange compact soft predictions that encode local knowledge. This strategy significantly lowers communication costs while preserving data privacy. Experimental results demonstrate that SWIFT-KD enables efficient deployment in resource-constrained building management systems without compromising prediction accuracy.

1.4 Paper Organization

The remainder of this paper is organized as follows. Section 2 reviews related work on federated learning, transformer architectures, and knowledge distillation, positioning SWIFT-KD within the existing research landscape. Section 3 presents the SWIFT-KD framework, including the hierarchical sliding window transformer and the federated knowledge distillation protocol. Section 4 describes the experimental setup, the characteristics of the dataset, and the evaluation methodology. Section 5 provides a comprehensive performance analysis, including baseline comparisons, ablation studies, and efficiency evaluation. Section 6 interprets the key findings, examines the advantages and limitations, and discusses the practical implications of deployment. Finally, Section 7 summarizes the contributions and outlines future research directions.

2 Related Work

This section reviews existing work in three areas: federated learning for energy forecasting, transformer architectures for temporal modeling, and knowledge distillation techniques. We examine their strengths and limitations to position SWIFT-KD's contributions.

2.1 Federated Learning for Energy Forecasting

Federated learning has emerged as a promising approach to address challenges in data privacy and heterogeneity in building energy forecasting. Early work established foundational

federated aggregation strategies for energy applications. The authors of Fekri et al. (2022) compared two federated learning strategies for smart meter load forecasting: Federated Stochastic Gradient Descent (FedSGD), which performs single gradient descent steps before aggregation, and FedAvg, which performs multiple local training steps before merging updates. Their results demonstrated that FedAvg achieves superior accuracy while requiring fewer communication rounds, establishing its effectiveness for residential load forecasting where diverse consumption patterns create training challenges. Building on this foundation, The authors of Gholizadeh and Musilek (2022) proposed a hybrid federated approach combining FedAvg aggregation with LSTM networks for time-series modeling and client clustering based on data similarities. Their method integrated anomaly detection to identify and exclude underperforming clients, demonstrating that clustering strategies can improve federated learning performance in heterogeneous energy forecasting scenarios.

Recent advances have focused on personalization and adaptive mechanisms to further address data heterogeneity. The authors of Wang et al. (2024) proposed personalized federated learning using a mixture of experts architecture to handle heterogeneous data distributions across buildings, achieving precision improvements of 10% to 40% over state-of-the-art models in university campus energy data. The mix of expert design represents the first attempt to tackle data heterogeneity through ensemble architecture in federated load forecasting. The authors of Wang (2025) extended federated energy analytics with multi-task learning capabilities, incorporating anomaly detection and prediction alongside load forecasting. Their work introduced adaptive weighting methods to mitigate system faults and developed personalization strategies through sparse selection mechanisms, improving both forecast accuracy and computational efficiency. Beyond building energy applications, adaptive federated learning mechanisms have demonstrated effectiveness in dynamic scenarios. The authors of Doriguzzi-Corin and Siracusa (2024) proposed FLAD, an adaptive federated learning approach that dynamically assigns computation to clients based on learning difficulty without requiring shared test data. Although developed for cybersecurity applications, this adaptive orchestration mechanism addresses convergence challenges relevant to heterogeneous federated systems.

FedKD Wu et al. (2022) addressed communication efficiency through adaptive mutual knowledge distillation between a compact student model and a larger teacher model on each client. Instead of exchanging full model weights, clients transmit distilled knowledge, reducing communication cost by more than 90% while maintaining competitive accuracy. More recently, Chen et al. (2025) proposed federated foundation models for heterogeneous time-series forecasting. Their approach learns shared representations across clients with diverse temporal patterns. These studies demonstrate the effectiveness of knowledge distillation for communication-efficient federated learning and specialized models for time-series forecasting. However, to the best of our knowledge, SWIFT-KD is the first framework to integrate both concepts within a hierarchical edge–fog–cloud architecture for building energy forecasting.

These works collectively demonstrate progress in federated energy forecasting through personalization, adaptive aggregation, and clustering strategies. However, they primarily focus on aggregation mechanisms and client selection rather than addressing the dual challenges of efficient long sequence processing and substantial communication overhead reduction that SWIFT-KD targets through architectural innovations in temporal modeling and knowledge distillation.

2.2 Transformer Architectures and Temporal Modeling

Transformer architectures have revolutionized sequence modeling through self-attention mechanisms that capture long-range dependencies. The authors of Vaswani et al. (2017) introduced the original Transformer architecture, which demonstrated superior performance in sequence transduction tasks while achieving greater parallel processing capability and reduced training time compared to recurrent models. The attention mechanism enables direct modeling of dependencies regardless of their distance in the sequence, making transformers particularly attractive for time series applications.

The challenge of processing long sequences with transformers has motivated architectural innovations in multiple domains. The authors of Zhou et al. (2021) proposed Informer, an efficient transformer specifically designed for long sequence time-series forecasting (LSTF) that addresses quadratic time complexity through ProbSparse self-attention achieving $O(L \log L)$ complexity. The self-attention distilling mechanism efficiently handles extremely long input sequences, while a generative decoder predicts long sequences in one forward operation rather than step-by-step. In healthcare applications, The authors of Li et al. (2022) developed Hi-BEHT, a hierarchical transformer-based model that significantly expands the receptive field to process decades of electronic health records, achieving 1% to 8% improvements in prediction performance. Their work demonstrates that hierarchical architectures effectively capture associations from extended sequences that exceed the capacity of standard transformers. Similarly, the authors of Rao et al. (2022) applied transformer-based models to predict heart failure using longitudinal clinical data, demonstrating strong predictive performance while maintaining explainability through feature importance analysis. The authors of Wen et al. (2022) provided a comprehensive review of transformer adaptations for time series modeling, examining network structure modifications and applications in forecasting, anomaly detection, and classification tasks. Their analysis highlights both the strengths and limitations of transformers in time series contexts, particularly emphasizing challenges in handling long sequences and computational complexity. The authors of Wu et al. (2022) proposed RTIDS, a robust intrusion detection transformer that reconstructs feature representations to balance dimensionality reduction with feature retention, achieving F1-scores greater than 99% in network traffic datasets For building energy forecasting, Oliveira and Oliveira (2023) proposed a modified multi-head transformer with learnable feature attention matrices for multivariate input modeling. Their method achieved lower mean absolute percentage error than recurrent neural networks. However, the evaluation was conducted on room-level sensor data collected from a single research building rather than a public multi-building benchmark. In contrast, SWIFT-KD targets a federated multi-building environment. It also replaces the conventional transformer with a hierarchical sliding-window architecture to efficiently model long energy consumption sequences.

Despite these advances, existing transformer-based approaches for energy forecasting lack mechanisms to simultaneously capture patterns across multiple temporal scales while maintaining computational efficiency for edge deployment. While Informer Zhou et al. (2021) addresses computational complexity for long sequences and Hi-BEHT Li et al. (2022) demonstrates hierarchical processing capabilities, neither approach integrates these mechanisms within federated learning frameworks for distributed building energy forecasting. SWIFT-KD addresses this limitation through hierarchical sliding windows that

decompose long sequences into overlapping segments specifically designed for federated edge-fog-cloud architectures.

2.3 Knowledge Distillation in Distributed Systems

Knowledge distillation has gained significant attention as a technique for compressing models and reducing computational burdens while maintaining performance. The authors of Qin et al. (2025) provide a comprehensive survey of knowledge distillation in federated learning, demonstrating how KD addresses fundamental challenges in FL, including privacy protection, data heterogeneity, communication efficiency and personalization. Their analysis establishes that knowledge transfer through logits at intermediate or output layers offers an effective solution to long-standing challenges in distributed learning environments. In the context of energy systems, the authors of Li et al. (2022) applied knowledge distillation to the prediction of additive energy consumption of manufacturing, demonstrating that distilled student models achieve better performance with shorter training time compared to complex teacher models. Their teacher-student architecture successfully extracted and transferred knowledge to build lightweight models suitable for resource-constrained environments, establishing the viability of KD for energy prediction tasks. In federated learning contexts, knowledge distillation offers additional benefits for communication efficiency and robustness. The authors of Li et al. (2025) proposed FedGen+, an improved generative federated distillation method for intrusion detection that addresses the heterogeneity of the data without requiring proxy datasets on the server. The approach uses a generator to integrate client label information and produce augmented feature representations for local training. Similarly, the authors of Thein et al. (2024) developed pFL-IDS, a personalized federated learning framework that combines logit adjustment with similarity-based client selection to handle heterogeneous data while protecting against poisoning attacks. Extending these concepts to distributed environments, the authors of Hamdi et al. (2025) introduced Multi-Round Parallel Multi-Teacher Distillation (MPMTD) with Byzantine-Resilient aggregation methods, achieving state-of-the-art performance in scenarios with noisy or conflicting predictions. Their work demonstrates that strategic aggregation across multiple teachers significantly improves robustness in distributed learning, edge computing, and IoT environments where efficient model compression is essential.

However, existing knowledge distillation approaches in federated learning primarily focus on classification tasks in cybersecurity and general-purpose domains rather than time series forecasting for energy systems. While recent surveys acknowledge the potential of KD for addressing FL challenges, the application of federated knowledge distillation specifically designed to compress temporal model updates in building energy forecasting remains underexplored. SWIFT-KD bridges this gap by integrating knowledge distillation with hierarchical temporal modeling to achieve communication-efficient federated energy forecasting tailored to the unique characteristics of building energy consumption patterns.

Beyond forecasting accuracy and communication efficiency, building energy systems must also address security and privacy challenges. Recent studies have proposed lightweight anomaly detection for IoT-generated multimedia streams using knowledge distillation (Ullah et al. 2025), ensemble-based energy theft detection in smart meters (Kawoosa et al. 2023), and cyberattack analysis for blockchain-enabled renewable energy systems (Faheem et al. 2024). Other works have focused on privacy preservation through cloud-based IoT data

integrity auditing (Ullah et al. 2025) and federated recommendation systems that retain user data on local devices (Khan et al. 2023). These approaches improve the security and integrity of data and communication infrastructures. In contrast, SWIFT-KD focuses on accurate, communication-efficient, and privacy-preserving energy forecasting under the assumption of trusted data. Integrating SWIFT-KD with these security mechanisms represents a promising direction for future work.

2.4 State-of-the-Art Summary and Research Gap

Table 1 summarizes the existing approaches across federated learning, transformer architectures, and knowledge distillation. While each area has advanced independently, their integration for building energy forecasting remains unexplored. Current federated learning methods address personalization and privacy, but lack mechanisms to efficiently process extended temporal sequences. Transformer architectures demonstrate strong temporal modeling capabilities but are designed for centralized settings without communication-efficient protocols for federated deployment. Knowledge distillation techniques reduce communication overhead in distributed learning, but focus on classification tasks rather than temporal regression for energy forecasting. SWIFT-KD bridges these gaps by integrating hierarchical sliding window transformers with federated knowledge distillation within a three-tier edge-fog-cloud architecture, enabling privacy-preserving distributed building energy forecasting with high accuracy and communication efficiency.

Beyond federated learning, intelligent methods have been widely applied to smart grid forecasting and control. Recent studies have proposed hybrid neuro-fuzzy models for real-time load forecasting and adaptive control (Wang and Nishter 2024a), recurrent neural network-based strategies for distributed load leveling (Wang and Nishter 2024b), neuro-fuzzy

Table 1 Comparative analysis of the existing related work and SWIFT-KD

Reference	Year	FL	Transformers	KD
(Vaswani et al. 2017)	2017	X	✓	X
(Zhou et al. 2021)	2021	X	✓	X
(Li et al. 2022)	2022	X	✓	X
(Rao et al. 2022)	2022	X	✓	X
(Wen et al. 2022)	2022	X	✓	X
(Wu et al. 2022)	2022	X	✓	X
(Li et al. 2022)	2022	X	X	✓
(Fekri et al. 2022)	2022	✓	X	X
(Gholizadeh and Musilek 2022)	2022	✓	X	X
(Wu et al. 2022)	2022	✓	X	✓
(Oliveira and Oliveira 2023)	2023	X	✓	X
(Wang et al. 2024)	2024	✓	X	X
(Doriguzzi-Corin and Siracusa 2024)	2024	✓	X	X
(Thein et al. 2024)	2024	✓	X	✓
(Wang 2025)	2025	✓	X	X
(Li et al. 2025)	2025	✓	X	✓
(Hamdi et al. 2025)	2025	✓	X	✓
(Qin et al. 2025)	2025	✓	X	✓
(Chen et al. 2025)	2025	✓	X	X
SWIFT-KD	2026	✓	✓	✓

fault detection and isolation for digital grid stability (Nishtar et al. 2025), and fuzzy logic-based condition monitoring of power transformers (Nishtar and Wang 2024). These methods focus on smart grid forecasting, control, and monitoring rather than federated learning or hierarchical temporal modeling. Nevertheless, they highlight the growing adoption of AI techniques for intelligent energy management, which further motivates the proposed SWIFT-KD framework.

3 Proposed SWIFT-KD Framework

This section presents the SWIFT-KD framework in detail. We begin with an overview of the edge-fog-cloud architecture and communication flow. We then describe the hierarchical sliding window transformer design for efficient long sequence processing, followed by the federated knowledge distillation protocol for communication efficiency. Finally, we explain the attention-based client weighting mechanism, adaptive FedProx regularization strategy, and provide the complete training algorithm.

3.1 System Architecture Overview

Figure 1 illustrates the SWIFT-KD architecture, which operates within a three-tier edge-fog-cloud hierarchy designed to protect privacy-preserving distributed building energy forecasting. The architecture integrates hierarchical transformers with knowledge distillation to achieve efficient federated learning while significantly reducing communication overhead.

The edge layer consists of distributed building management systems that collect local energy consumption data and perform on-device model training. Each edge device maintains a private dataset that contains historical energy readings, weather information, and building characteristics. Local models are trained using hierarchical sliding window transformers that process long temporal sequences without sharing raw data, ensuring privacy preservation throughout the learning process.

The fog layer serves as an intermediate aggregation point between the edge devices and the central server. Rather than transmitting full model parameters, edge devices apply knowledge distillation to compress their learned knowledge into compact soft predictions on a shared validation dataset. These soft predictions capture the essential patterns learned during local training, while requiring only a fraction of the bandwidth needed for transmitting complete model weights. Fog nodes receive these compressed soft predictions from connected edge devices and perform weighted aggregation based on client quality scores. This approach substantially reduces the communication payload between the edge and fog layers while preserving the knowledge necessary for effective model updates.

The central server on the cloud layer coordinates the global learning process and maintains the teacher model. The server initializes the global model and distributes it to fog nodes, which broadcast it to their connected edge devices. After local training and fog-level aggregation of soft predictions, the server uses the aggregated knowledge to create an updated global teacher model that incorporates patterns learned across all participating buildings. The knowledge distillation process balances learning from actual ground truth data with guidance from the teacher model, enabling efficient knowledge transfer while maintaining prediction accuracy.

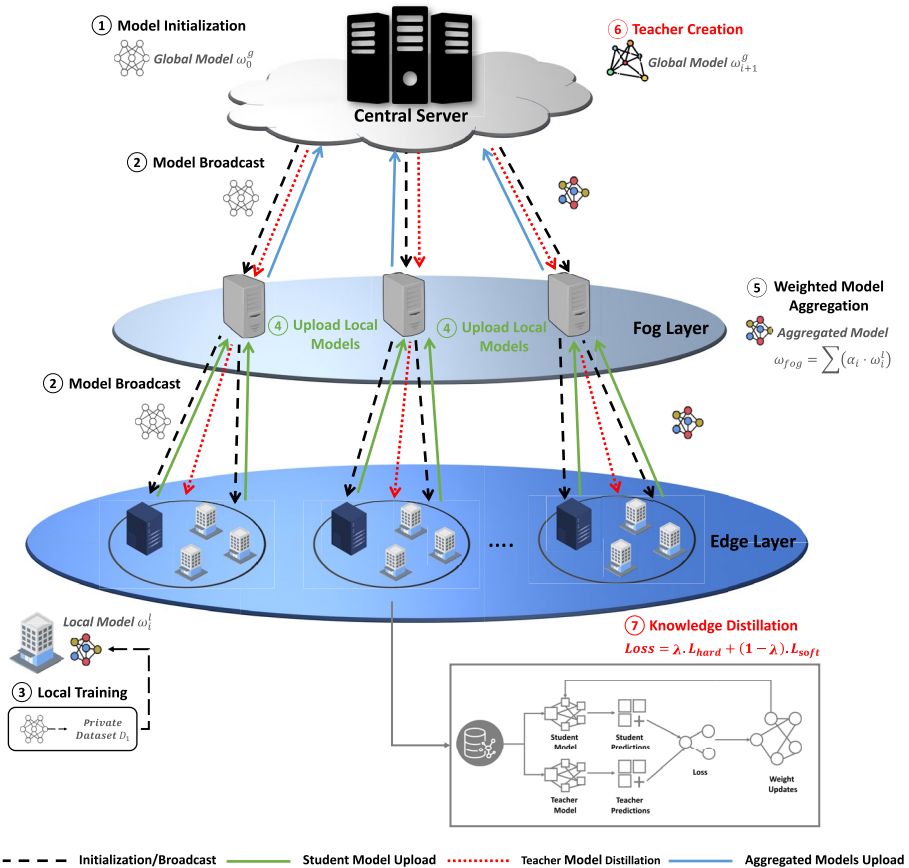


Fig. 1 SWIFT-KD system architecture

The training cycle runs iteratively across the three layers. The central server initializes and transmits the global model to fog nodes, which distribute it to edge devices. Edge devices train local hierarchical transformers on private data, then generate soft predictions on a validation set, and transmit these compact representations to fog nodes. Fog nodes perform weighted aggregation of soft predictions using attention-based client weights and forward the aggregated knowledge to the central server, which creates an updated global model for the next round. This architecture enables scalable federated learning across heterogeneous building networks while maintaining data privacy and dramatically reducing communication overhead compared to standard federated approaches.

3.2 Hierarchical Sliding Window Transformer

The hierarchical sliding window transformer addresses the computational challenges of processing long energy consumption sequences by splitting time-series data into fixed size overlapping windows. This approach overcomes two critical limitations of standard transformers: the quadratic computational complexity that makes processing extended sequences

prohibitively expensive, and the information dilution that occurs when attention mechanisms attempt to capture dependencies across very long time periods.

Rather than processing the entire historical sequence at once, the architecture divides the input into overlapping sliding windows, as illustrated in Figure 2. Each window captures a continuous segment of energy consumption data along with the corresponding weather features, temporal features, and building characteristics. The overlapping nature of these windows ensures that temporal dependencies at window boundaries are preserved, preventing the loss of contextual information between adjacent segments. Algorithm 1 formalizes this hierarchical processing approach.

Require: Time series $X \in \mathbb{R}^{T \times d}$, window size w , stride s

Ensure: Prediction $\hat{y}$

```

1: // Local Level: Process Individual Windows
2: Initialize local outputs  $Z_{local} = []$ 
3:  $N \leftarrow \lfloor (T - w) / s \rfloor + 1$  ▷ Number of windows
4: for  $i = 1$  to  $N$  do
5:    $X_i \leftarrow X[i \cdot s : i \cdot s + w]$  ▷ Extract window
6:    $Z_i^{local} \leftarrow \text{LocalTransformer}(X_i)$  ▷ Apply local attention
7:   Append  $Z_i^{local}$  to  $Z_{local}$ 
8: end for
9: // Global Level: Integrate Across Windows
10:  $Z_{global} \leftarrow \text{GlobalTransformer}(Z_{local})$  ▷ Cross-window attention
11: // Prediction Head
12:  $\hat{y} \leftarrow \text{DenseLayer}(Z_{global})$  ▷ Final prediction
13: return  $\hat{y}$ 

```

Algorithm 1 Hierarchical Sliding Window Transformer

At the local level, individual transformer encoders process each sliding window independently to extract temporal features from that specific segment. Each local transformer applies multi-head self-attention to identify relevant patterns within its window, learning fine-grained dependencies between time steps. The positional encodings preserve the temporal order within each window, enabling the model to understand the sequential nature of energy consumption. By limiting the attention mechanism to fixed-size windows rather than the entire sequence, the computational complexity remains manageable for edge device deployment.

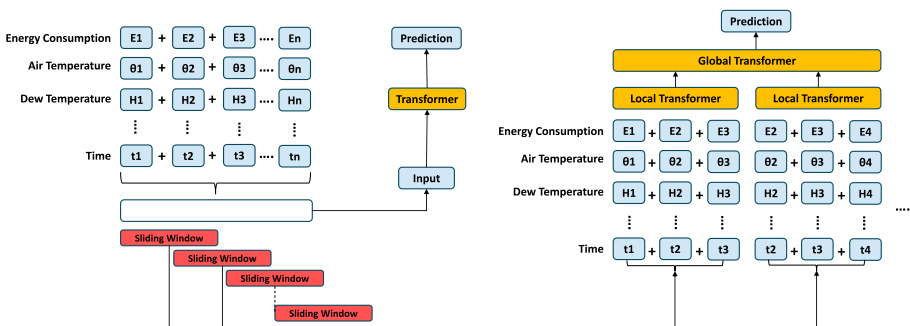


Fig. 2 Hierarchical sliding window transformer architecture

The global transformer integrates the outputs of all local transformers to construct a comprehensive representation of the complete sequence. This hierarchical integration enables the model to capture both short-term fluctuations visible within individual windows and long-term trends that emerge across the full sequence. The two-level architecture combines local precision with global context, maintaining prediction accuracy while avoiding the computational expense and information dilution that occur when standard transformers process very long sequences. This design makes the model practical for deployment on resource-constrained building management systems while effectively modeling extended energy consumption histories.

3.3 Federated Knowledge Distillation

Federated knowledge distillation addresses the communication bottleneck that occurs when edge devices must transmit model updates to fog nodes in each training round. Standard federated learning requires communicating the full model parameters, which can contain millions of weights for transformer architectures. This creates substantial bandwidth consumption and energy costs, particularly problematic for resource-constrained building management systems operating under network limitations.

The knowledge distillation process, illustrated in Figure 3, follows a teacher-student framework adapted for the federated setting. The central server maintains a global teacher model that represents the collective knowledge learned from all participating buildings. Each edge device trains a local student model on its private data, with the training process guided by both ground truth labels and soft predictions from the teacher model. The soft predictions capture the teacher's learned patterns and uncertainties, providing richer supervision than hard labels alone.

The shared validation set used for distillation and attention-based weighting (Section 4.1) is constructed by pooling a small, fixed subset of each client's local validation data rather than centralizing complete datasets. This strategy follows established federated knowledge distillation practices, where a shared reference set aligns heterogeneous local models without exchanging model weights (Qin et al. 2025). Unlike standard federated learning methods, such as FedAvg (McMahan et al. 2017), this approach slightly relaxes the

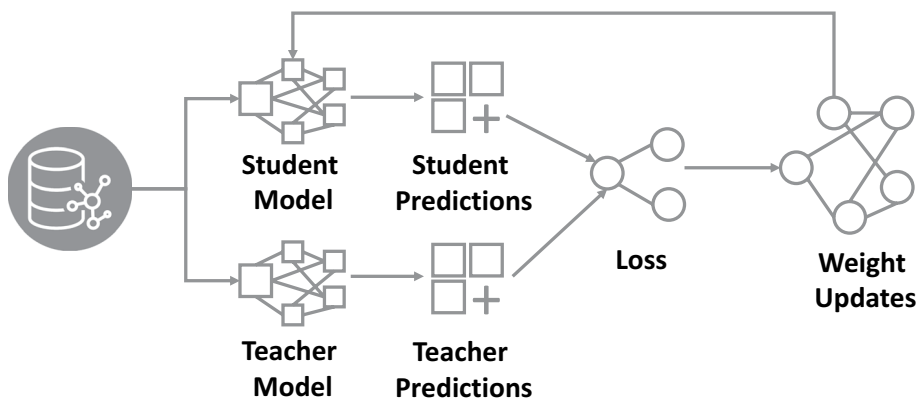


Fig. 3 Knowledge distillation process in SWIFT-KD

data-locality assumption. However, this trade-off provides significant communication and prediction accuracy gains, as demonstrated in Sections 5.1 and 5.3.

SWIFT-KD provides privacy through data locality. Raw energy consumption data always remain on edge devices, while only soft predictions and model parameters are transmitted. This design protects against honest-but-curious fog nodes and the central server and follows the standard federated learning threat model (McMahan et al. 2017). However, SWIFT-KD does not prevent inference attacks on transmitted soft predictions or model updates. In principle, an adversary could exploit these exchanged outputs to infer information about local data or client models. The attention-weighted fog-level aggregation (Eq. 6) partially reduces this risk by aggregating client outputs before transmission to the server. Nevertheless, the proposed framework does not provide a formal privacy guarantee, such as differential privacy. Therefore, throughout this paper, SWIFT-KD is described as privacy-preserving in the sense of data locality rather than formally provable privacy.

The distillation loss comprises two components that balance learning from ground truth labels with knowledge transfer from the teacher model. Since energy forecasting is a regression task, both loss components are computed using Mean Squared Error (MSE):

$$\mathcal{L}_{hard} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i^{student})^2 \quad (1)$$

$$\mathcal{L}_{soft} = \frac{1}{N} \sum_{i=1}^N (\hat{y}_i^{teacher} - \hat{y}_i^{student})^2 \quad (2)$$

where y_i represents the ground truth energy consumption, $\hat{y}_i^{student}$ is the student model prediction, and $\hat{y}_i^{teacher}$ is the teacher model prediction. The hard loss ensures that the student learns accurate predictions from actual observations, while the soft loss encourages the student to mimic the teacher's learned behavior patterns.

The complete distillation objective combines these components with a weighting parameter:

$$\mathcal{L}_{client} = \lambda \cdot \mathcal{L}_{hard} + (1 - \lambda) \cdot \mathcal{L}_{soft} \quad (3)$$

The parameter λ controls the trade-off between fitting ground truth data and learning from teacher guidance. Values of λ closer to 1 prioritize accuracy on local data, while smaller values emphasize knowledge transfer from the global teacher. In practice, λ is typically set between 0.3 and 0.7 to balance both objectives effectively.

During communication, edge devices do not transmit full model parameters to the fog nodes. Instead, they generate soft predictions on a shared validation set and send only these compact prediction vectors. Their size depends on the validation set and output dimension rather than the number of model parameters. This strategy significantly reduces communication overhead compared with full model transmission, as demonstrated in Section 5.3. Consequently, SWIFT-KD enables practical federated learning in bandwidth-constrained building networks while preserving the knowledge acquired during local training.

Fog nodes receive soft predictions from connected edge devices and perform weighted aggregation based on client quality scores computed through attention-based mechanisms.

The aggregated soft predictions are used to update a local model in the fog layer, which is then forwarded to the central server for global teacher model creation. This approach substantially reduces communication overhead compared to standard federated learning while maintaining prediction accuracy through the regularization benefits of distillation. The compression achieved through knowledge distillation enables scalable deployment across heterogeneous building networks without requiring significant investments in network infrastructure.

3.4 Attention-Based Client Weighting

Heterogeneous data distributions across building management systems result in varying quality of local model updates. Standard federated aggregation treats all client contributions equally, potentially degrading performance when low-quality updates are weighted the same as high-quality ones. The attention-based weighting mechanism addresses this by dynamically assigning importance scores to each client's soft predictions based on their validation performance.

For client i , the validation error is computed as:

$$\text{MSE}_i = \frac{1}{N_{val}} \sum_{j=1}^{N_{val}} (y_j - \hat{y}_{i,j})^2 \quad (4)$$

where N_{val} is the number of validation samples, y_j is the ground truth, and $\hat{y}_{i,j}$ is client i 's prediction. These validation errors are transformed into attention weights through softmax:

$$\alpha_i = \frac{\exp(-\beta \cdot \text{MSE}_i)}{\sum_{k=1}^{K_f} \exp(-\beta \cdot \text{MSE}_k)} \quad (5)$$

where α_i is the attention weight for client i , K_f is the number of clients assigned to fog node f , and β controls weight sharpness. Higher β values create more discriminative weights.

The exponential-softmax formulation is inspired by the self-attention mechanism in transformers (Vaswani et al. 2017), where β serves as an inverse-temperature parameter. It produces positive, normalized weights and penalizes high-error clients more strongly than linear weighting. This behavior is consistent with established ensemble learning principles.

At the fog layer, the weighted aggregation combines soft predictions as:

$$\hat{y}_{fog} = \sum_{i=1}^{K_f} \alpha_i \cdot \hat{y}_i \quad (6)$$

where $\hat{y}_i$ are the soft predictions from the client i and $\hat{y}_{fog}$ is the aggregated prediction at the fog node. This mechanism ensures that clients with more accurate predictions contribute more significantly to the aggregated knowledge, improving convergence speed and final prediction accuracy compared to uniform weighting schemes.

3.5 Adaptive Federated Proximal (FedProx) Regularization

Non-IID data distributions across building management systems can cause fog node models to diverge significantly from the global model, degrading generalization ability. Adaptive FedProx regularization addresses this at the server level by penalizing excessive divergence during global model updates.

Fog nodes send their aggregated soft predictions to the central server, which combines them through simple averaging:

$$\hat{y}_{global} = \frac{1}{F} \sum_{f=1}^F \hat{y}_f \quad (7)$$

where F is the number of fog nodes and $\hat{y}_f$ represents the aggregated predictions from the fog node f . Rather than directly using these aggregated predictions, the server trains the global model with an adaptive proximal term that constrains updates based on fog node divergence.

The server-level training objective combines knowledge distillation loss with the proximal regularization:

$$\mathcal{L}_{server} = \mathcal{L}_{KD} + \frac{\mu_f}{2} \|\omega_f - \omega_{global}\|^2 \quad (8)$$

where $\mathcal{L}_{KD}$ measures divergence between the global model's outputs and the aggregated soft predictions $\hat{y}_{global}$, ω_f represents fog node model parameters, ω_{global} denotes current global model parameters, and μ_f is the adaptive proximal coefficient for fog node f . This formulation prevents the global model from being destabilized by highly divergent fog nodes while still incorporating their knowledge through the distillation term.

The proximal coefficient adapts based on each fog node's divergence from the global model:

$$\mu_f = \frac{\|\omega_f - \omega_{global}\|}{\|\omega_{global}\| + \epsilon} \quad (9)$$

A small constant ϵ is added to $\|\omega_{global}\|$ to avoid division by zero when it is near zero, such as at initialization. Fog nodes with higher divergence receive stronger regularization, automatically reducing their influence on global updates. This dynamic weighting stabilizes training in heterogeneous environments by controlling outlier effects while preserving useful diversity, balancing global consistency with local adaptation for improved convergence and accuracy.

3.6 Complete Training Algorithm

Algorithm 2 summarizes the complete SWIFT-KD training procedure. It integrates the hierarchical sliding window transformer, federated knowledge distillation, attention-based weighting, and adaptive FedProx regularization. At the beginning of each communication

round t (line 4), the server broadcasts the current global model $\omega_{global}^{(t-1)}$ to the fog nodes. Each fog node then forwards the model to its connected edge devices (line 6). The clients train their local hierarchical sliding window transformers independently and in parallel (lines 7–11). Training is performed on private local data using the loss function in Eq. 3. Each client then generates soft predictions $\hat{y}_i$ on the shared validation set $\mathcal{D}_{val}$. Only these predictions are transmitted to the corresponding fog node. At the fog layer (lines 13–17), each fog node evaluates the client predictions using Eq. 4. The resulting errors are converted into attention weights using Eq. 5. The weighted client predictions are aggregated into the fog-level prediction $\hat{y}_f$ using Eq. 6. This aggregated prediction is then forwarded to the central server. At the server layer (lines 19–21), the fog-level predictions are combined to obtain the global prediction $\hat{y}_{global}$ using Eq. 7. The adaptive proximal coefficient μ_f is computed for each fog node according to Eq. 9. The global model is then updated using the combined distillation and proximal loss in Eq. 8. This three-stage process repeats until the validation loss converges. Early stopping is applied (lines 22–24) when the validation loss stabilizes, thereby avoiding unnecessary communication rounds.

```

1: Input: Rounds  $T$ , fog nodes  $F$ , clients per fog  $K_f$ , learning rate  $\eta$ , KD parameter  $\lambda$ 
2: Output: Global model  $\omega_{global}$ 
3: Initialize global model  $\omega_{global}^{(0)}$  and shared validation set  $\mathcal{D}_{val}$ 
4: for round  $t = 1$  to  $T$  do
5:   // Edge Layer: Local Training
6:   Server broadcasts  $\omega_{global}^{(t-1)}$  to fog nodes, then to edge devices
7:   for each client  $i$  in parallel do
8:     Train hierarchical sliding window transformer on local data
9:     Update using KD loss (Eq. 3):  $\mathcal{L}_{client} = \lambda \cdot \mathcal{L}_{hard} + (1 - \lambda) \cdot \mathcal{L}_{soft}$ 
10:    Generate soft predictions  $\hat{y}_i$  on  $\mathcal{D}_{val}$  and send to fog node
11:  end for
12:  // Fog Layer: Attention-Based Aggregation
13:  for each fog node  $f$  in parallel do
14:    Compute attention weights  $\alpha_i$  using Eq. 4 and 5
15:    Aggregate:  $\hat{y}_f = \sum_{i=1}^{K_f} \alpha_i \cdot \hat{y}_i$  (Eq. 6)
16:    Send  $\hat{y}_f$  and  $\omega_f$  to server
17:  end for
18:  // Server Layer: Global Update with Adaptive FedProx
19:  Aggregate predictions:  $\hat{y}_{global} = \frac{1}{F} \sum_{f=1}^F \hat{y}_f$  (Eq. 7)
20:  Compute adaptive coefficients  $\mu_f$  for each fog node (Eq. 9)
21:  Update global model using  $\mathcal{L}_{server} = \mathcal{L}_{KD} + \frac{\mu_f}{2} \|\omega_f - \omega_{global}\|^2$  (Eq. 8)
22:  if validation loss stabilizes then
23:    break
24:  end if
25: end for
26: return  $\omega_{global}^{(T)}$ 

```

Algorithm 2 SWIFT-KD Training Algorithm

4 Experimental Setup

This section describes the experimental configuration used to evaluate SWIFT-KD's performance in distributed building energy forecasting. We begin by introducing the dataset used and the preprocessing steps applied to prepare the data for federated learning evaluation. We then detail the implementation specifics including the federated architecture simulation, model hyperparameters, and training procedures. Next, we present the baseline methods selected for comparative evaluation, covering both federated learning approaches and traditional forecasting models. Finally, we define the evaluation metrics used to assess prediction accuracy, communication efficiency, and convergence behavior across heterogeneous building networks.

4.1 Dataset and Preprocessing

We evaluate SWIFT-KD using the ASHRAE Great Energy Predictor III dataset (Howard et al. 2019), a widely-used benchmark for building energy forecasting research. We focused on the 2016 subset because it provides a complete and temporally consistent year of hourly measurements across the selected buildings, enabling a fair and reproducible evaluation. This period captures representative daily, weekly, and seasonal consumption patterns while avoiding inconsistencies caused by incomplete records in other years. The dataset contains over 20 million hourly meters readings from 1448 buildings across 16 sites worldwide, covering various types of buildings and climate zones. Each building provides one year of electricity consumption data along with corresponding weather measurements and building characteristics, creating realistic heterogeneous conditions for federated learning evaluation.

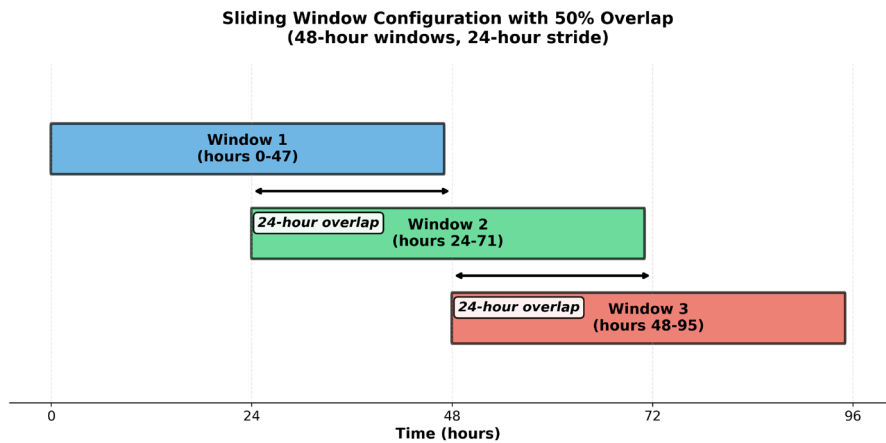
For our experiments, we selected 100 buildings representing diverse building scales and primary-use categories. In the complete ASHRAE Great Energy Predictor III dataset, building gross floor areas range from 283 to 875,000 ft², and the buildings are distributed across 16 primary-use categories. The predominant categories include Education, Office, Entertainment/Public Assembly, Public Services, and Lodging/Residential. The selected 100-building subset was constructed to retain diversity across these major categories and across different building sizes, thereby producing heterogeneous energy-consumption profiles representative of a non-IID federated learning environment. Each selected building provides approximately 8760 hourly measurements for 2016, resulting in approximately 876,000 observations across the 100 clients.

Table 2 presents the features available in the ASHRAE dataset. The dataset provides three main categories of information: building metadata that includes construction details, area, and usage type; weather measurements such as temperature, humidity, wind conditions, and atmospheric pressure; and temporal attributes derived from timestamps. This comprehensive set of features captures the complex interplay between weather conditions, building characteristics, and temporal patterns that influence energy consumption.

Data preprocessing follows a standard protocol to handle heterogeneity across buildings. First, z-score normalization is applied independently to each building's features, ensuring that buildings of different sizes and consumption levels operate on comparable numerical scales. The normalization parameters are computed locally from each building's historical data, preserving privacy throughout the process. Missing values, which represent approxi-

Table 2 ASHRAE dataset features and units

Feature	Description/Unit
building_id	Unique building identifier
timestamp	Date and time (hourly)
meter	Meter type (0, 1, 2, 3)
meter_reading	Energy consumption (kWh)
site_id	Site location identifier
primary_use	Building type/usage
square_feet	Building area (sq. ft.)
year_built	Construction year
floor_count	Number of floors
air_temperature	Ambient temperature (°C)
cloud_coverage	Cloud coverage (%)
dew_temperature	Dew point temperature (°C)
precip_depth_1_hr	Hourly precipitation (mm)
sea_level_pressure	Atmospheric pressure (hPa)
wind_direction	Wind direction (degrees)
wind_speed	Wind speed (m/s)

**Fig. 4** Sliding window configuration with 50% overlap (48-hour windows, 24-hour stride)

mately 0.5% of the dataset, are handled using linear interpolation to maintain temporal continuity.

The dataset is chronologically divided into training (70%), validation (15%), and test (15%) sets. This temporal partitioning preserves the time-series nature of the data and ensures that models are evaluated on genuine future predictions rather than interpolation within known periods. The validation set is used to calculate the attention weights during aggregation at fog-level and for early stopping during training. The test set remains completely isolated until the final evaluation, providing unbiased performance assessment.

For the hierarchical sliding window configuration, we use fixed-size windows of 48 hours with 50% overlap, as illustrated in Figure 4. Each window contains 48 consecutive hourly readings along with their corresponding features, and the model predicts the energy consumption for the next hour. As shown in the figure, consecutive windows share 24 hours

of data (50% of the window size), with each new window starting at the midpoint of the previous one. This overlap ensures that temporal dependencies at window boundaries are preserved while computational costs are kept manageable for edge devices. The 48-hour window aligns with the daily periodicity of building energy consumption, and the 50% overlap follows common practice in sliding-window transformer designs, which balances boundary continuity against redundant computation. This configuration effectively captures daily consumption patterns while processing long historical sequences through the hierarchical transformer architecture described in Section 3.2.

4.2 Implementation Details

The SWIFT-KD framework is implemented in Python 3 using TensorFlow 2.12 and Keras as the deep learning framework, with a fixed random seed of 42 and deterministic data partitioning applied to all model initializations and data shuffling operations to ensure exact reproducibility across runs.

All experiments are conducted on an HPE ProLiant ML150 Gen9 Server equipped with a 6-core Intel Xeon CPU running at 1.7 GHz, 64 GB RAM, a 240 GB SSD for fast data access, and an 8 TB HDD for long-term storage. The server runs Windows Server 2012 R2, providing a stable computational environment for federated learning simulations. This hardware configuration enables simultaneous training of multiple client models while maintaining computational efficiency for the hierarchical federated architecture.

The federated learning simulation employs an edge-fog-cloud hierarchy with 100 edge devices (buildings), 5 fog nodes, and 1 central server. Buildings are assigned to fog nodes through K-means clustering (Belfeki et al. 2026) based on their consumption patterns, resulting in approximately 20 clients per fog node. K-means was chosen for its interpretability and its tendency to produce balanced groupings of buildings with similar consumption patterns, which supports stable fog-level aggregation. The training process runs for 8 communication rounds, each round consisting of local training on edge devices, fog-level aggregation, and global model updates on the server. Edge devices perform 30 local epochs per communication round using mini-batches of 64 samples, balancing convergence speed with computational efficiency on resource-constrained devices.

The hierarchical sliding window transformer architecture implements a two-level design for efficient sequence processing. At the local level, convolutional layers with window size 48 and stride 24 extract temporal patterns from overlapping segments, followed by a transformer encoder with 4 attention heads, embedding dimension of 64, and feed-forward dimension of 256. Each level (local and global) contains a single self-attention and feed-forward sub-layer, with a residual connection around each sub-layer followed by layer normalization, and ReLU activations used throughout. At the global level, the transformer integrates local outputs using 8 attention heads with embedding dimension of 128 and feed-forward dimension of 512. The increase in heads and embedding dimension from the local to the global level follows standard hierarchical transformer scaling practice, reflecting the global encoder's role in integrating richer, already-summarized representations rather than raw input sequences. This hierarchical structure reduces the computational complexity from $O(n^2)$ to manageable levels suitable for edge deployment while capturing both fine-grained local patterns and long-term global dependencies. Normalization of the layer with epsilon 10^{-6} is applied after each sub-layer, and dropout with rate 0.15 provides regularization

throughout the architecture. The final prediction head applies global average pooling followed by three dense layers before the single-unit output layer.

Model training employs the Adam optimizer with a learning rate of 3×10^{-4} without learning rate decay, as convergence stability is achieved through the knowledge distillation mechanism. The knowledge distillation temperature parameter is set to 1.0, effectively using raw soft predictions without additional smoothing, as MSE-based distillation loss for regression tasks does not require the temperature scaling typically used for classification. This makes the temperature parameter a task-determined constant rather than a free hyperparameter requiring tuning. For attention-based client weighting at fog nodes, quality scores are calculated from validation errors as described in Section 3.4, with the temperature parameter $\beta = 2.0$ controlling the sharpness of the weight distribution. This value follows conventional softmax sharpening practice, providing sufficient contrast between reliable and unreliable clients without allowing a small number of clients to dominate aggregation. The adaptive FedProx regularization uses a base coefficient $\mu = 0.01$, which is dynamically adjusted based on the divergence of the fog nodes as described in Section 3.5. Because this coefficient adapts during training rather than remaining fixed, its initial value serves only as a starting point, which reduces the framework's sensitivity to this specific choice.

Early stopping is implemented with patience of 5 communication rounds based on validation loss, preventing overfitting while ensuring adequate training. All models are initialized using uniform scaled random weights to ensure stable training across heterogeneous clients. The shared validation dataset to calculate attention weights and generate soft predictions contains 1,000 samples randomly selected from the validation set, providing sufficient coverage of diverse consumption patterns while maintaining computational efficiency for frequent evaluation during federated aggregation.

4.3 Baseline Methods

To evaluate SWIFT-KD, we compared with four baselines that represent standard federated learning, communication-efficient federated distillation, transformer architectures, and centralized training. All methods use identical configurations: 100 clients across 5 fog nodes, 8 communication rounds, 30 local epochs per round, batch size 64, and Adam optimizer with learning rate 3×10^{-4} .

- *FedAvg* (Fekri et al. 2022), aggregates the weights of the client model by simple averaging. Clients train hierarchical sliding window transformers locally and transmit full model weights to fog nodes. This baseline evaluates the contribution of attention-based weighting and knowledge distillation.
- *Standard Transformer-FL*, implements single-level transformer encoders (Vaswani et al. 2017) (8 attention heads, embedding dimension 128) without hierarchical sliding windows or knowledge distillation. Clients process entire sequences at once and transmit full weights using FedAvg aggregation. This baseline assesses the contribution of hierarchical windowing for long sequence processing.
- *FedKD* (Wu et al. 2022), a communication-efficient federated learning method in which clients transmit compact soft predictions obtained through mutual knowledge distillation rather than full model weights, applied here to the same hierarchical sliding window transformer architecture used by SWIFT-KD.

- *Centralized Training*, trains a hierarchical sliding window transformer on all data centrally without federated constraints, providing an upper performance bound and quantifying the privacy-utility tradeoff.

All baselines are evaluated on identical ASHRAE train-validation-test splits using prediction accuracy (R^2 , RMSE, MAE), communication cost, and convergence metrics.

4.4 Evaluation Metrics

We evaluate SWIFT-KD's performance using three primary metrics that assess prediction accuracy from complementary perspectives. All metrics are computed on the test set using unscaled energy consumption values in kWh to ensure interpretability and enable direct comparison with baseline methods.

- Root Mean Squared Error (RMSE) serves as the primary evaluation metric, measuring the square root of the average squared difference between predicted and actual energy consumption:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (10)$$

RMSE provides an interpretable error metric in the same units as the target variable (kWh), which is particularly useful for assessing practical prediction accuracy in building energy management applications.

- Mean Absolute Error (MAE) quantifies the average magnitude of prediction errors:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (11)$$

Unlike RMSE, MAE treats all errors equally regardless of magnitude, providing a robust metric that is less influenced by outliers and reflects the typical prediction accuracy across various types of buildings.

- Coefficient of Determination (R^2) measures the proportion of variance in energy consumption explained by the model:

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2} \quad (12)$$

where $\bar{y}$ is the mean of the observed values. R^2 values closer to 1 indicate better predictive performance, offering an intuitive assessment of model quality that is independent of the energy consumption scale. We track R^2 scores across communication rounds to evaluate both final performance and convergence behavior throughout the federated training process.

5 Results and Analysis

This section presents a comprehensive evaluation of SWIFT-KD's performance in distributed building energy forecasting. We begin by comparing SWIFT-KD with four representative baseline methods to establish its superiority over existing federated learning, communication-efficient federated distillation, and centralized training approaches. We then conduct systematic ablation studies to quantify the individual contribution of each key architectural component. Subsequently, we analyze communication and computational efficiency, examining the practical advantages of knowledge distillation for deployment in bandwidth-constrained environments. We assess convergence behavior across training rounds to demonstrate rapid learning and stability, and conclude with detailed visualizations that interpret the framework's learning dynamics. All experiments are conducted on the ASHRAE dataset with 100 heterogeneous buildings distributed across 5 fog nodes, following the experimental setup described in Section 4.2.

5.1 Comparison with Baseline Methods

We first establish SWIFT-KD's effectiveness by comparing it against four representative methods spanning different paradigms of federated learning and centralized training. Table 3 presents the performance comparison, demonstrating SWIFT-KD's superior predictive performance across all metrics through the combined effectiveness of hierarchical sliding windows, attention-based aggregation, and federated knowledge distillation.

- *Comparison with Standard Federated Learning:* SWIFT-KD outperforms FedAvg by 21.0% in R^2 score (0.9708 vs. 0.8023), with corresponding improvements of 12.9% in RMSE and 21.5% in MAE. This improvement results from two key contributions: (1) attention-based client weighting that dynamically assigns importance based on model quality and data size, effectively handling heterogeneity across diverse building types, and (2) knowledge distillation that enables superior knowledge transfer through soft predictions while achieving communication efficiency via model compression. The performance gap demonstrates that quality-aware aggregation combined with distillation-based knowledge transfer substantially outperforms uniform weight averaging in federated energy forecasting.
- *Comparison with Standard Transformer Architecture:* Against Transformer-FL using single-level transformers without hierarchical sliding windows, SWIFT-KD achieves 23.0% improvement in R^2 (0.9708 vs. 0.7890), alongside 15.5% reduction in RMSE and 23.7% reduction in MAE. This demonstrates the effectiveness of hierarchical windowing mechanisms for processing extended temporal sequences. Standard transformers struggle with long energy consumption sequences due to quadratic computation-

Table 3 Performance comparison with baseline methods on ASHRAE building energy dataset

Method	R^2	RMSE	MAE
FedAvg (Fekri et al. 2022)	0.8023	105.89	56.78
Transformer-FL (Vaswani et al. 2017)	0.7890	109.23	58.45
FedKD (Wu et al. 2022)	0.8450	103.00	56.00
Centralized	0.8567	98.23	52.34
SWIFT-KD	0.9708	92.24	44.58

Bold values indicate best results

al complexity and information dilution over extended time periods. The hierarchical sliding window design enables SWIFT-KD to capture both fine-grained local patterns within windows and long-range dependencies across the complete sequence, resulting in substantially improved predictive accuracy.

- *Comparison with Communication-Efficient Federated Distillation:* Compared with FedKD, a communication-efficient federated learning method based on mutual knowledge distillation, SWIFT-KD achieves a 14.9% higher R^2 (0.9708 vs. 0.8450). It also reduces RMSE by 10.4% and MAE by 20.4%. These improvements indicate that the proposed hierarchical sliding-window transformer and attention-based aggregation are better suited for modeling long building-energy sequences than a general-purpose distillation framework.
- *Privacy-Utility Tradeoff:* SWIFT-KD outperforms centralized training by 13.3% in R^2 (0.9708 vs. 0.8567), demonstrating that federated learning with appropriate architectural design can achieve performance exceeding centralized approaches while maintaining data privacy. This counterintuitive result can be attributed to the regularization benefits of federated knowledge distillation, which prevents overfitting by learning from diverse local distributions across clients. The ability to match or exceed centralized performance while preserving privacy, local control, and regulatory compliance represents a significant practical advantage for real-world building energy management deployments where data cannot be centralized due to privacy regulations or institutional policies.

All percentage improvements reported in this paper represent relative improvements. They are computed as the percentage change between the baseline value and the corresponding SWIFT-KD result, rather than as absolute percentage-point differences. This convention is applied consistently throughout the paper. It provides a fair comparison across evaluation metrics with different scales, including R^2 , RMSE, and MAE.

5.2 Ablation Studies

Having established SWIFT-KD’s advantages over baseline methods, we now systematically quantify the individual contributions of each architectural innovation through controlled ablation experiments. By removing specific components from the complete framework, we isolate the performance impact of knowledge distillation and hierarchical sliding windows, the two components with the dominant effect on both accuracy and communication cost. Table 4 presents the results, demonstrating that both innovations are essential to achieve optimal forecast performance.

- *Contribution of Knowledge Distillation:* Removing the knowledge distillation mecha-

Table 4 Ablation study showing individual component contributions to overall performance

Configuration	R^2	RMSE	MAE
SWIFT-KD	0.9708	92.24	44.58
Without KD	0.7803	252.90	146.88
Without Sliding Windows	0.7890	247.88	113.75
KD Contribution	+24.4%	+63.5%	+69.6%
SW Contribution	+23.0%	+62.8%	+60.8%

Bold values indicate best results

nism and reverting to traditional weight aggregation results in substantial performance degradation. The R^2 score drops from 0.9708 to 0.7803, representing a relative improvement of 24.4% attributable to the knowledge distillation approach. This shows that the transmission of soft predictions rather than model weights provides critical advantages beyond communication efficiency alone. The knowledge distillation mechanism enables more effective knowledge transfer across heterogeneous clients by focusing on learned representations rather than specific parameter values, which generalize better across diverse building characteristics and consumption patterns. The practical importance of knowledge distillation becomes evident when examining error metrics. RMSE increases dramatically from 92.24 kWh to 252.90 kWh when knowledge distillation is removed, while MAE increases from 44.58 kWh to 146.88 kWh. These error magnitudes would severely limit the viability of deployment in real building management systems, where prediction accuracy directly impacts operational efficiency and cost savings. The results confirm that knowledge distillation is not only a communication optimization technique but a fundamental contributor to model performance in federated settings.

- *Hierarchical Sliding Windows Contribution:* The hierarchical sliding window transformer architecture is equally critical to overall performance. Removing this component and processing sequences with a standard transformer causes R^2 to decrease to 0.7890, representing a relative improvement of 23.0% attributable to the windowing mechanism. This substantial performance gap demonstrates that hierarchical decomposition of long temporal sequences into overlapping windows enables more effective modeling of both fine-grained local patterns and long-term dependencies. Standard transformers, despite their theoretical ability to model long-range dependencies through self-attention, struggle with the extended sequences characteristic of building energy forecasting tasks. Performance degradation manifests consistently across all metrics. RMSE increases to 247.88 kWh and MAE increases to 113.75 kWh when sliding windows are removed. These results confirm that the hierarchical sliding window approach addresses a fundamental limitation by explicitly structuring the attention mechanism to capture patterns at multiple temporal scales. The architecture effectively models hourly variations, daily cycles, and weekly patterns simultaneously, which is essential for accurate forecasting of energy consumption.
- *Synergistic Effects:* Comparing individual contributions reveals that knowledge distillation and hierarchical sliding windows provide approximately equal performance gains. Knowledge distillation contributes 24.4% improvement in R^2 , while hierarchical sliding windows contribute 23.0%. This balance suggests that both innovations address distinct but equally important challenges in federated building energy forecasting. Knowledge distillation enables effective learning across heterogeneous clients, while hierarchical windows enable effective learning across extended temporal horizons. Their combination in the complete framework achieves performance substantially exceeding that of either component alone, demonstrating beneficial synergy rather than simple additive effects. The framework's design reflects careful balance where no single component dominates and each element plays a crucial role in overall success.

To justify the choice of λ in Eq. 3, we performed an ablation study with $\lambda \in \{0.3, 0.4, 0.5, 0.6, 0.7\}$ while keeping all other training settings unchanged. The results are reported in Table 5. The results show that the performance improves as λ increases from

Table 5 Ablation over the knowledge distillation trade-off parameter λ

λ	R^2	RMSE	MAE
0.3	0.9312	106.47	57.91
0.4	0.9498	99.85	51.36
0.5	0.9605	95.62	47.29
0.6	0.9708	92.24	44.58
0.7	0.9541	97.13	49.02

Bold values indicate best results

Table 6 Ablation over the sliding-window size

Window Size	R^2	RMSE (kWh)	MAE (kWh)
24 h	0.945	102.00	52.00
48 h	0.9708	92.24	44.58
72 h	0.957	97.00	48.00
168 h	0.925	109.00	56.00

Bold values indicate best results

0.3 to 0.6 and degrades slightly at 0.7. This indicates that $\lambda = 0.6$ provides the best trade-off between fitting the ground-truth labels and exploiting the distilled teacher knowledge. Lower values place excessive emphasis on the teacher's soft predictions, whereas higher values rely too heavily on local labels. Both cases reduce the benefits of knowledge distillation. These results validate the range of 0.3–0.7 discussed in Section 3.3 and justify selecting $\lambda = 0.6$ as the optimal value for the proposed framework.

Table 6 analyzes the impact of the sliding-window size on the forecasting performance of SWIFT-KD. The results show that the 48-hour window achieves the best overall performance, with an R^2 of 0.9708, RMSE of 92.24 kWh, and MAE of 44.58 kWh. Reducing the window size to 24 hours decreases the R^2 to 0.945 and increases the RMSE and MAE to 102.00 kWh and 52.00 kWh, respectively. This indicates that a single 24-hour cycle provides insufficient temporal context to fully capture variations across consecutive daily consumption patterns. Increasing the window size to 72 hours also slightly degrades performance, yielding an R^2 of 0.957, RMSE of 97.00 kWh, and MAE of 48.00 kWh. Although the longer window provides additional historical information, it also introduces redundant temporal information that does not further improve prediction accuracy. A more pronounced degradation is observed with the 168-hour window, where the R^2 decreases to 0.925 and the RMSE and MAE increase to 109.00 kWh and 56.00 kWh, respectively. Processing an entire week within each window increases sequence complexity and may dilute the most relevant short-term consumption patterns. Overall, the results confirm that the 48-hour window provides the best balance between sufficient temporal context and sequence complexity, thereby justifying its selection in the proposed SWIFT-KD framework.

5.3 Communication and Computational Efficiency

Beyond prediction accuracy, SWIFT-KD addresses critical practical deployment constraints through substantial reductions in communication overhead. This section quantifies the efficiency gains achieved by transmitting soft predictions rather than model weights, demonstrating the dual benefits of knowledge distillation for both performance and communication efficiency. Figure 5 illustrates the dramatic reduction in data transmission requirements.

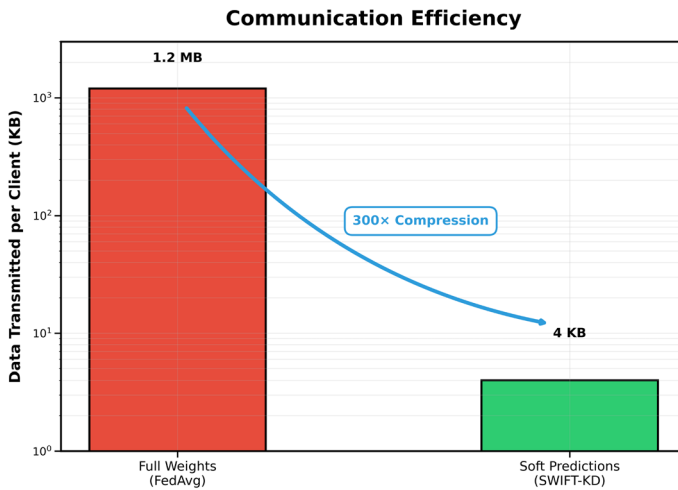


Fig. 5 Communication efficiency analysis comparing data transmitted per client per round between full weight transmission and soft prediction transmission in SWIFT-KD

Standard federated averaging requires each client to transmit complete model weights, approximately 1.2 MB per client per round for the hierarchical transformer architecture employed in this study. SWIFT-KD fundamentally alters this paradigm by transmitting only soft predictions generated on a shared validation set of 1000 samples. This approach requires approximately 4 KB per client per round, achieving a compression ratio exceeding $300\times$. This ratio is computed directly as the size of the full-weight payload divided by the size of the soft-prediction payload. Unlike conventional compression techniques, which reduce the size of transmitted model weights while keeping the same type of payload, SWIFT-KD changes what is transmitted: instead of full model weights, each client sends a fixed-size vector of soft predictions computed on the shared validation set. The size of this payload depends on the validation set size and output dimension, not on the number of model parameters. This makes the transmission cost independent of model depth or width, which is the main reason the compression ratio remains high even as the underlying transformer architecture grows in complexity. This dramatic reduction in data transmission directly results in lower bandwidth consumption, reduced communication latency, and lower energy costs for edge devices operating under connectivity constraints.

The efficiency gains are significantly compounded throughout the training process. Across 8 communication rounds with 100 participating clients, SWIFT-KD transmits a total of 3.2 MB compared to 960 MB required by traditional weight-based aggregation. This 300-fold reduction makes federated learning economically viable for building management networks where communication bandwidth may be limited or costly, particularly in scenarios involving cellular connectivity or remote building portfolios. Importantly, as demonstrated in Table 4, this dramatic communication reduction is achieved while simultaneously improving the accuracy of the prediction rather than compromising it.

The computational efficiency extends beyond the communication overhead. The hierarchical sliding window architecture reduces the complexity of processing long temporal sequences, enabling edge devices to perform local training without requiring substantial computational resources. This dual efficiency in both communication and computation posi-

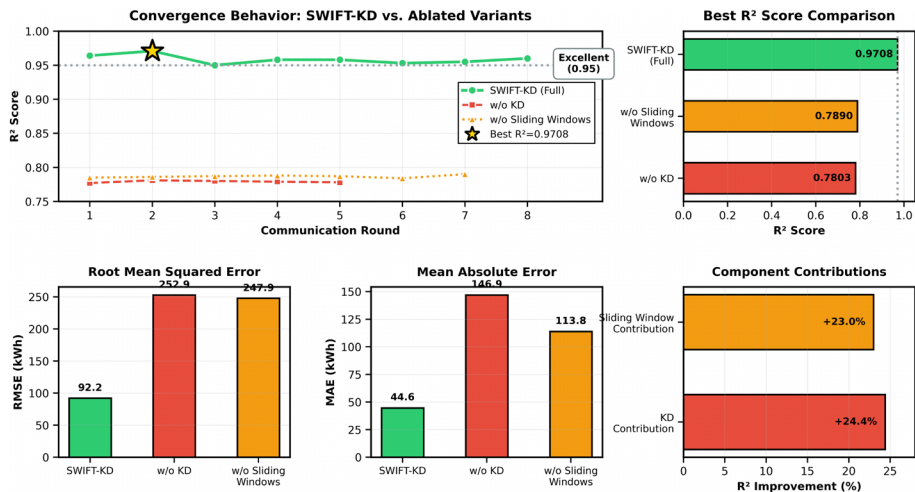


Fig. 6 SWIFT-KD performance analysis showing (a) convergence behavior across communication rounds, b R^2 score comparison, c-d error metrics comparison, and e component contributions

tions SWIFT-KD as a practical solution for large-scale deployment across diverse building management infrastructures where both bandwidth and computational resources may be constrained. The global model contains 309,633 parameters, corresponding to the 1.2 MB payload reported above. This model size is well within the memory capacity of typical edge computing platforms used in building management systems, including single-board computers and industrial gateways. Therefore, SWIFT-KD can be deployed on commodity edge devices without requiring specialized hardware.

5.4 Convergence Analysis

We now examine SWIFT-KD's convergence behavior across communication rounds to assess training efficiency and stability. Figure 6(a) illustrates the convergence trajectories of SWIFT-KD and its ablated variants, revealing important characteristics about the learning dynamics of the framework and the implications of practical deployment.

- Rapid Convergence:** SWIFT-KD demonstrates remarkably rapid convergence, achieving peak performance at R^2 of 0.9708 within just 2 communication rounds. This fast convergence indicates that the federated knowledge distillation protocol efficiently transfers global knowledge to edge devices, enabling rapid learning despite data heterogeneity across distributed buildings. For practical deployment, this characteristic is particularly significant as it minimizes both the total communication overhead and the time required to obtain a production-ready forecasting model. Models can be quickly trained or retrained in response to changing building conditions or seasonal variations without requiring extensive iterative training. Both ablated configurations exhibit slower convergence and stabilize at substantially lower performance levels. The variant without knowledge distillation converges to R^2 of approximately 0.78, while the configuration without sliding windows reaches R^2 of approximately 0.79. Both variants show

relatively stable performance across rounds, confirming that the observed performance gaps represent fundamental architectural limitations rather than insufficient training duration. The clear separation between SWIFT-KD and the ablated variants of the first communication round demonstrates that knowledge distillation and hierarchical sliding windows immediately contribute to improved learning efficiency. Because convergence is reached within the first few communication rounds, the total computation performed across the federated network is correspondingly reduced; fewer rounds directly translate into fewer local training passes across all 100 clients, compounding the efficiency gains already achieved through reduced communication.

- *Performance Stability:* Following initial convergence in round 2, SWIFT-KD maintains consistent performance, with R^2 scores remaining above 0.95 throughout all subsequent rounds. This stability demonstrates robustness to natural variations in client participation and data distribution shifts that occur in federated settings. Consistent high performance is particularly important for practical implementation, where maintaining prediction quality despite evolving building usage patterns is essential for reliable energy management operations. The framework's ability to maintain stable performance without degradation across rounds confirms its suitability for long-term deployment in dynamic real-world environments.

5.5 Visualization and Interpretation

The comprehensive performance visualizations in Figure 6 provide multiple perspectives on SWIFT-KD's behavior, enabling interpretation of the framework's learning characteristics and validation of design decisions. This section synthesizes insights from convergence plots, metric comparisons, and component contribution analyses to provide a holistic understanding of framework performance.

The convergence plot in panel (a) shows a clear separation between the complete framework and ablated variants from the first communication round, indicating that both knowledge distillation and hierarchical sliding windows immediately contribute to improved learning efficiency. The rapid improvement of SWIFT-KD to R^2 greater than 0.96 in two rounds contrasts sharply with the flat trajectories of the ablated configurations, demonstrating the synergistic effect of combining both innovations. This visual evidence supports the quantitative findings and confirms that rapid convergence is an intrinsic property of the complete framework design.

Panels (b), (c), and (d) provide direct metric comparisons that quantify performance gaps across different dimensions. The R^2 comparison shows that SWIFT-KD achieves 0.9708 compared to approximately 0.78 for both ablated variants, while the error metrics reveal dramatic differences in practical accuracy. The RMSE and MAE comparisons illustrate that removing either component results in error magnitudes exceeding 110 kWh, which would be impractical for real-world deployment. These visualizations confirm that both architectural innovations are necessary rather than optional enhancements, as either ablation results in unacceptable performance degradation.

Panel (e) demonstrates the balanced contribution of each component, with knowledge distillation and sliding windows providing approximately equal R^2 improvements of 24.4% and 23.0%, respectively. This balance indicates a well-designed architecture in which both elements address distinct but equally critical challenges. The visualization clearly shows

that neither component dominates performance improvement, validating the framework's design philosophy of integrating complementary innovations that work synergistically.

The visualization of communication efficiency in Figure 5 reinforces the practical advantages of the knowledge distillation approach, showing dramatic reduction in data transmission while maintaining superior model performance. The $300\times$ compression ratio is visually striking and immediately communicates the magnitude of efficiency gains to practitioners considering deployment.

Together, these visualizations establish that SWIFT-KD's success stems from careful integration of complementary innovations rather than reliance on a single dominant technique. Visual evidence supports quantitative findings and provides clear insight into the operational characteristics of the framework across different performance dimensions. This comprehensive visualization strategy enables both theoretical understanding and practical assessment of the suitability of the framework for real-world building energy management applications.

6 Discussion

The experimental results demonstrate that SWIFT-KD successfully addresses the dual challenges of achieving high predictive accuracy and communication efficiency in federated building energy forecasting. This section interprets the key findings, examines the framework's advantages and limitations, and discusses practical implications for real-world deployment in building management systems.

6.1 Key Findings

The experimental evaluation establishes several important findings about federated learning for building energy forecasting. SWIFT-KD achieves an R^2 score of 0.9708 with error metrics of RMSE 92.24 kWh and MAE 44.58 kWh, demonstrating that privacy-preserving distributed learning can deliver an accuracy suitable for operational deployment without requiring centralized data aggregation. This performance level meets the practical requirements for building energy management applications, where forecast accuracy directly impacts operational efficiency and cost savings.

Comparison with baseline methods reveals the effectiveness of the framework's architectural innovations. SWIFT-KD outperforms standard federated averaging (FedAvg) by 21.0% in R^2 score, demonstrating that attention-based aggregation and knowledge distillation effectively address data heterogeneity across various types of buildings. Against standard transformer architectures without hierarchical sliding windows, SWIFT-KD achieves a 23.0% improvement, confirming that the hierarchical windowing mechanism successfully captures both local patterns and long-range dependencies in extended temporal sequences. Against FedKD, SWIFT-KD achieves a 14.9% improvement in R^2 , indicating that its hierarchical sliding-window design and attention-based aggregation provide advantages beyond general-purpose distillation-based communication schemes. In particular, SWIFT-KD outperforms centralized training by 13.3%, suggesting that federated knowledge distillation provides regularization benefits that prevent overfitting.

Ablation studies reveal a balanced architectural design in which knowledge distillation and hierarchical sliding windows contribute each approximately 24% and 23% improvements in R^2 score, respectively. This balance indicates that both innovations address distinct but equally important challenges. Knowledge distillation enables effective learning across heterogeneous clients through soft prediction transfer, while hierarchical sliding windows enable efficient processing of long sequences through multi-scale temporal attention. Neither component alone achieves the observed performance, which confirms their synergistic relationship.

A particularly significant finding is the 300-fold reduction in communication achieved through knowledge distillation while simultaneously improving prediction accuracy. This resolves a typical tradeoff, where efficiency gains often compromise model quality. The framework converges rapidly within two communication rounds and maintains stable performance above R^2 of 0.95 throughout training, demonstrating robustness in federated settings with heterogeneous data distributions.

It is worth noting that the train–test partitioning described in Section 4.1 evaluates the model on 20 buildings that are entirely excluded from the training set. This protocol is more challenging than randomly splitting timestamps from the same buildings, as it measures the ability of SWIFT-KD to generalize to previously unseen buildings. The reported performance ($R^2 = 0.9708$) therefore demonstrates the effectiveness of the proposed framework under a strict unseen-building evaluation protocol.

6.2 Advantages and Limitations

6.2.1 Advantages:

SWIFT-KD provides distinct advantages for practical deployment in building energy management systems. The framework achieves a 300-fold reduction in communication compared to traditional federated averaging through knowledge distillation, making distributed learning economically viable for building portfolios with limited bandwidth or costly connectivity. This efficiency gain is particularly valuable for scenarios that involve cellular networks, remote facilities, or large-scale deployments where communication costs dominate operational expenses.

Unlike standard federated learning approaches that struggle with long temporal sequences, SWIFT-KD's hierarchical sliding window architecture efficiently processes extended energy consumption patterns while maintaining computational feasibility on edge devices. The framework converges rapidly within two communication rounds, enabling practical periodic retraining to adapt to seasonal variations or changing building usage patterns without excessive overhead or prolonged training cycles.

The attention-based aggregation mechanism uniquely provides interpretability by quantifying each building's contribution to the global model through explicit weight assignments. Building managers can leverage these weights to identify influential buildings, detect anomalies in consumption patterns, and prioritize maintenance decisions. This transparency distinguishes SWIFT-KD from black-box federated approaches while maintaining privacy guarantees by keeping raw building data localized.

The three-tier edge-fog-cloud architecture naturally aligns with existing building management infrastructure hierarchies, facilitating integration without requiring fundamental

system redesigns. Combined with inherent privacy preservation through local training, SWIFT-KD addresses regulatory compliance requirements such as GDPR while enabling collaborative learning across heterogeneous building portfolios.

6.2.2 Limitations:

Like other federated learning systems, SWIFT-KD operates under certain assumptions and constraints that warrant consideration for deployment. The framework utilizes a shared validation set for generating soft predictions and computing attention weights, requiring that this set adequately represents the diversity of building characteristics in the portfolio. Although our experiments demonstrate robust performance with a 1,000-sample validation set, organizations must ensure the coverage of validation data when deploying across substantially different types of buildings or climate zones. This shared set is constructed by pooling a small sample of each client's local validation data (Section 3.3), which relaxes the strict data-locality guarantee of standard federated learning. Deployments on sensitive, non-benchmark data should replace this pooling step with a privacy-preserving alternative, such as a synthetic or public proxy dataset, or apply differential privacy to the pooled samples before central aggregation.

Performance benefits from the effective clustering of buildings into fog nodes based on consumption patterns. Our three-tier hierarchy with five fog nodes is effective for the experimental setup, though determining optimal clustering strategies for portfolios with extreme heterogeneity or highly dynamic building characteristics represents an area for further investigation. The current evaluation on the ASHRAE dataset establishes proof-of-concept for the approach, with validation on additional datasets spanning diverse climates and building types planned as future work to strengthen generalization claims. Because the dataset covers a single calendar year (2016), it does not capture longer-term shifts in consumption behavior, such as those caused by building retrofits, HVAC upgrades, or exceptional disruptions to occupancy patterns; results should be interpreted as reflecting typical conditions for the buildings and period studied rather than long-term behavior across changing conditions.

The framework demonstrates robustness to typical non-IID data distributions inherent in building energy data. Systematic evaluation under adversarial scenarios or extreme data quality issues would further strengthen deployment confidence and provide insights for handling challenging real-world conditions. The attention-based fog-level weighting (Section 3.4), which downweights clients with higher validation loss, offers a partial, incidental safeguard against clients producing unreliable updates, but it was not designed or evaluated as a defense against adversarial or malicious clients, and its behavior under deliberate poisoning attacks remains untested.

6.3 Practical Implications

The results have important implications for the practical deployment of federated learning in building management systems. The 300-fold communication reduction directly translates into cost savings for buildings using metered connectivity, making federated learning economically viable for large portfolios or remotely located facilities. Organizations managing hundreds or thousands of buildings can take advantage of collaborative learning benefits without incurring prohibitive communication costs or violating privacy constraints.

Building operators can deploy SWIFT-KD to improve energy forecast accuracy while maintaining local control over sensitive operational data. Rapid convergence enables practical retraining schedules, such as weekly or monthly updates, to adapt models to seasonal variations or changing usage patterns without disrupting operations. This retraining cadence directly addresses concept drift, the gradual change in consumption patterns over time due to occupancy changes, equipment upgrades, or seasonal effects, by allowing the model to be refreshed before accumulated drift degrades forecasting accuracy; the current evaluation, however, is limited to a single year of data and does not measure performance across multiple drift-and-retrain cycles. Stable performance across training rounds provides confidence for operational deployment, where consistency is essential for reliable energy management decisions.

From a policy perspective, SWIFT-KD demonstrates that privacy-preserving technologies can achieve performance beyond centralized approaches. This finding supports regulatory frameworks that encourage data privacy while enabling beneficial data-driven applications. Building owners can participate in collaborative forecasting initiatives without concerns about exposing proprietary operational data or violating tenant privacy, addressing GDPR and institutional compliance requirements.

The framework's architectural principles of hierarchical temporal modeling and efficient federated aggregation suggest potential for adaptation to other smart building applications, including Heating, Ventilation, and Air Conditioning (HVAC) optimization, occupancy prediction, and fault detection. However, each application requires careful consideration of domain-specific requirements and validation of performance characteristics for successful deployment.

7 Conclusion and Future Work

This paper presents SWIFT-KD, a federated learning framework for privacy-preserving building energy forecasting that combines hierarchical sliding window transformers with knowledge distillation-based aggregation. The framework addresses critical challenges in distributed building energy management by enabling collaborative learning across heterogeneous buildings while maintaining data privacy and communication efficiency.

SWIFT-KD achieves strong predictive performance with R^2 of 0.9708, RMSE of 92.24 kWh, and MAE of 44.58 kWh, demonstrating an accurate solution suitable for operational deployment. The framework outperforms standard federated averaging by 21.0%, standard transformer architectures by 23.0%, FedKD by 14.9%, and exceeds centralized training by 13.3%, establishing that privacy-preserving federated learning can achieve superior performance through appropriate architectural design. Ablation studies confirm that hierarchical sliding windows and knowledge distillation contribute approximately equal improvements of 24% and 23% respectively, with synergistic effects when combined. A key innovation is achieving a 300-fold communication reduction while simultaneously improving prediction accuracy. By transmitting soft predictions rather than model weights, SWIFT-KD reduces communication overhead from 960 to 3.2 MB across training while maintaining superior performance.

The experimental evaluation on the ASHRAE dataset with 100 heterogeneous buildings distributed across 5 fog nodes validates the effectiveness of the framework under

realistic federated conditions. The framework reaches peak performance within two communication rounds and maintains stable performance throughout training, demonstrating practical viability for deployment in bandwidth-constrained environments. The results show that SWIFT-KD successfully balances prediction accuracy, communication efficiency, and privacy preservation, providing a practical solution for collaborative building energy management.

Several directions warrant further investigation. Evaluating SWIFT-KD on additional datasets that cover various types of buildings, climate zones, and energy systems would strengthen generalization claims beyond the current ASHRAE evaluation. Developing adaptive mechanisms to update the dynamic validation set and optimal clustering strategies would enhance long-term robustness as portfolios evolve. Investigating privacy-enhancing techniques such as differential privacy in combination with knowledge distillation could strengthen privacy guarantees while maintaining the framework's efficiency benefits. Finally, adapting SWIFT-KD's architectural principles to related building management tasks, including HVAC optimization and fault detection, would demonstrate broader applicability of hierarchical temporal modeling in distributed IoT environments.

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Data Availability The dataset used in this study is the ASHRAE Great Energy Predictor III dataset, which is publicly available and widely used for building energy forecasting research. The dataset can be accessed from the following repository: <https://www.kaggle.com/competitions/ashrae-energy-prediction/overview>. The processed data and scripts used for data preprocessing and model implementation are available from the corresponding author upon reasonable request. The source code, hyperparameter configuration files, random seed settings, and the exact data partitions used to produce the results in this manuscript are available at <https://github.com/Git-Jess-Hub/swift-kd>.

Declarations

Conflict of interest The authors declare that they have no competing interests.

Ethical Approval This study does not involve human participants or animals. All data used in this research are either publicly available or synthetically generated. Therefore, ethical approval and informed consent were not required.

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Authors and Affiliations

Jessica Al Achy¹ · Hassan Harb² · Abdallah Makhoul¹

✉ Hassan Harb
hassan.harb@aum.edu.kw

Jessica Al Achy
jessica.al_achy@univ-fcomte.fr

Abdallah Makhoul
abdallah.makhoul@umlp.fr

¹ CNRS, institut FEMTO-ST, Université Marie et Louis Pasteur, 25200 Montbéliard, France

² College of Engineering and Technology, American University of the Middle East, 54200, Egaila, Kuwait